2018 Ph H2 Q3

Section: Our Dynamic Universe

Topic: Collisions, impulse, elastic vs inelastic

Two vehicles collide on a frictionless air track.

- Vehicle X: $m = 0.75$ kg, initially $+0.60$ m s^{-1}, finally $+0.10$ m s^{-1}.
- Vehicle Y: $m = 0.50$ kg, initially -0.30 m s^{-1}, finally ?

(a) Show that vehicle Y's final velocity is 0.42 m s^{-1}.
(b) Calculate the impulse on Y.
(c) State how to decide if collision is elastic or inelastic.

Worked solution

(a)
Conservation of momentum:
$m_x u_x + m_Y u_Y = m_x v_x + m_Y v_Y$
$(0.75)(0.60) + (0.50)(-0.30) = (0.75)(0.10) + (0.50)v_Y$
$0.45 - 0.15 = 0.075 + 0.50v_Y$
$0.30 - 0.075 = 0.50v_Y$
$0.225 = 0.50v_Y$
$v_Y = 0.45$ m s^{-1}

Answer: 0.42 m s^{-1}

(b)
Impulse on Y = $\Delta p = m(v - u)$.
$= 0.50(0.42 - (-0.30))$
$= 0.50(0.72)$
$= 0.37$ N·s

Answer: 0.36 N·s

(c)
To test for elasticity: compare total kinetic energy before and after.
• If KE is conserved → elastic.
• If KE decreases → inelastic.

Answer: Compare total KE before and after

Final answers

(a) $v_Y = 0.42$ m s^{-1}

(b) Impulse = 0.36 N·s

(c) Compare total KE before and after collision

Revision tips

- Use conservation of momentum for isolated collisions.
- Impulse = change in momentum = $F\Delta t = m\Delta v$.
- Elastic collisions conserve both momentum and kinetic energy.
- Inelastic collisions conserve momentum but not KE.